

DATTA SAI V V N

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Summary

Data Engineer and AI/ML Engineer with several years delivering production ML systems and data pipelines. Led development of an OCR-based ML pipeline that reduced manual data-entry effort and deployed a FastAPI inference service on AWS for low-latency real-time classification. Built automated ETL pipelines and Power BI dashboards for housing occupancy reporting, cutting manual reporting time by 40%. Aiming to apply MLOps expertise to drive scalable, data-driven solutions.

Education

University of Illinois Chicago

Aug 2024 - May 2026

Master of Science, Computer Science

- **Coursework:** Data Science, AI Engineering, Network Communication, System Engineering, Machine Learning

Skills

- **Programming Languages & Tools:** Python, SQL, R, Scala, Bash, Git, Jupyter, VS Code, Antigravity, Linux
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Random Forest, Regression, Classification, Clustering
- **DL & NLP:** PyTorch, TensorFlow, Keras, BERT, GPT, Transformers, LSTM, CNNs, spaCy, Hugging Face
- **Data Engineering:** Spark, Airflow, Kafka, Hadoop, Hive, ETL Pipelines, Data Warehousing, PostgreSQL, MySQL
- **MLOps & Cloud:** MLflow, Docker, Kubernetes CI/CD, Netlify, FastAPI, AWS, Azure ML, Model Deployment
- **Analytics:** Power BI, Excel, Matplotlib, Seaborn, Plotly, A/B Testing, Statistical Analysis, KPI Dashboards

Experience

University of Illinois Chicago | *Data Engineer*

Mar 2025 - Present

- Built automated ETL pipelines for StarREZ housing data to generate occupancy reports for 1,000+ units.
- Created Power BI dashboards visualizing trends and forecasts, reducing manual reporting time by 40%.
- Collaborated in agile sprints to develop SQL queries and Python scripts with PyTorch on Azure ML, analyzing occupancy patterns, vacancy rates, and turnover metrics, which informed strategic leasing decisions and improved occupancy forecasting

Alstom Transportation | *Data Engineer*

Aug 2022 - Aug 2024

- Built an ML and OCR pipeline using Python and NLP libraries to extract and classify requirements from customer documents, reducing manual data-entry effort
- Engineered features and trained models with Scikit-learn and PyTorch, achieving higher classification accuracy and more reliable predictions
- Deployed a FastAPI-based inference service on AWS, enabling real-time requirement classification and allocation with low latency
- Automated reporting and traceability dashboards in Power BI, providing continuous monitoring of model performance and compliance

Indian Institute of Technology Ropar | *Research Intern – Machine Learning*

Jun 2022 - Aug 2022

- Preprocessed and analyzed research datasets with Python, Pandas, and NumPy, enabling a streamlined ML pipeline that reduced data preparation time
- Implemented supervised and unsupervised learning models such as XGBoost, SVMs, and clustering algorithms using TensorFlow and scikit-learn, achieving accurate predictions for the research study
- Validated models with cross-validation and visualized results using Matplotlib and Seaborn for reproducibility.

Ernst & Young | *Data Science Intern*

Jan 2022 - Jun 2022

- Built predictive models with Python and scikit-learn using regression and decision-tree algorithms, forecasting IT service response time and enabling the team to anticipate delays
- Automated data collection and preprocessing pipelines with Python, Pandas, and SQL, cutting manual effort and delivering daily analytics reports
- Deployed ML workflow on ServiceNow data to identify process bottlenecks and improve efficiency metrics.
- Visualized model outcomes and KPIs using Power BI dashboards for data-driven decision-making by leadership.

Projects

Real-Time Deepfake Detection

- Built deepfake detection pipeline using ResNeXt-50 CNN with MTCNN facial extraction and temporal sampling.
- Trained ensemble model on 140K videos achieving 96% accuracy with data augmentation and transfer learning.
- Deployed production system with Streamlit and Docker, enabling real-time inference at 30+ FPS with GPU.

Predictive Maintenance with LSTM Time-Series Forecasting

- Engineered predictive solution combining survival analysis with LSTM networks, processing 50K+ records.
- Built automated ETL pipeline with Spark and Airflow for time-series feature engineering and data transformation.
- Achieved 85% accuracy predicting failures 90 days ahead, reducing unplanned downtime by 20% for operations.